ELEC345 Tutorial 13/10/25 Milya 10803133

CHECKED 3/11/25

1. What 3 things causes signal loss in wires?

- ~~Material of wire~~
- ~~Length of wire~~
- ~~Skin effect~~

- Radiation
- Skin Effect
- Reflection

2. When do you start needing transmission lines?

~~when the distance between the transmission and receiver is too big~~

when physical length is 0.1 of the wavelength, λ

3. What is a transmission line?

~~wires made to transmit data efficiently based on requirements~~

2 parallel conductors

4. A 50Ω (Z_0) line is connected to an open circuit what is the reflection coefficient Γ

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$ and open circuit has ∞ resistance

$$\Gamma = \frac{\infty - 50}{\infty + 50} = 1$$

5. A 50Ω (Z_0) line is connected to a short circuit what is the reflection coefficient Γ

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$ and short circuit has 0 resistance

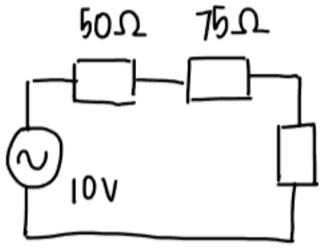
$$\Gamma = \frac{0 - 50}{0 + 50} = -1$$

6. A 50Ω line is connected to a matched load (50Ω), what is Γ ?

$$Z_0 = Z_L = 50\Omega$$

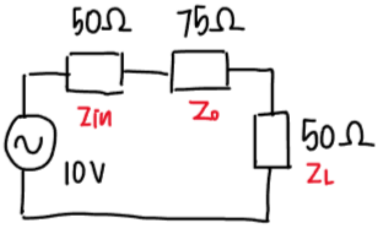
$$\Gamma = \frac{50 - 50}{50 + 50} = 0$$

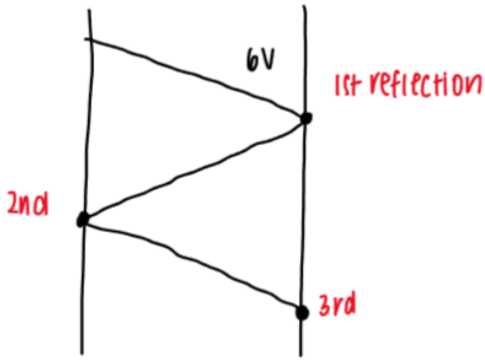
7. A signal generator with output R=50Ω is connected to a 75Ω line. How much voltage is delivered onto the line if its output is 10V?



$$V_{Z_0} = \frac{75}{50+75}(10) = 6V$$

8. If the line in previous question is connected to a 50Ω load. Calculate the size of the first 3 reflections on the line.





$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = -0.2$$

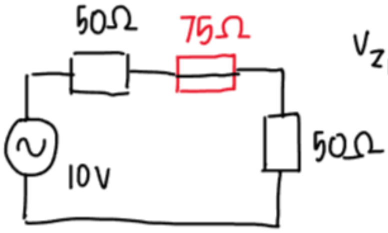
$$\Gamma_S = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = -0.2$$

1st reflection $= 6(-0.2) = -1.2V$

2nd reflection $= 6(-0.2)^2 = 0.24V$

3rd reflection $= 6(-0.2)^3 = -0.048V$

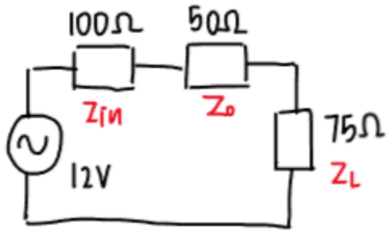
9. What voltage will the line settle to?

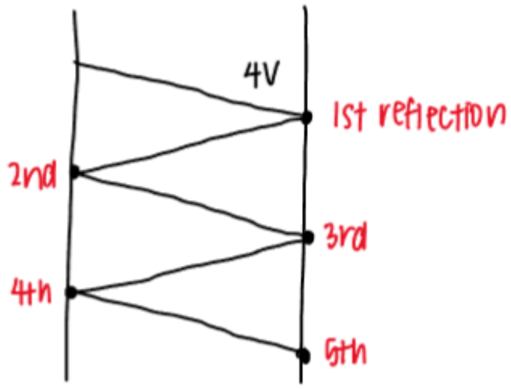


$$V_{Z_L} = \frac{50}{50+50}(10) = 5V$$

Line will settle to 5V

10. A signal generator with output R=100Ω is connected to a 50Ω line with a 75Ω load connected and voltage of12V. Calculate the first 5 reflections and find the settled voltage of the line.



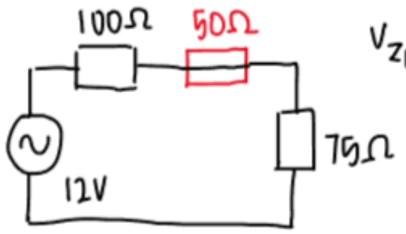


$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = 0.2$$

$$\Gamma_S = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = \frac{1}{3} = 0.33$$

$$V_{Z_0} = \frac{50}{100+50}(12) = 4V$$

1st reflection $= 4(0.2) = 0.8$
2nd reflection $= 4(0.2)(1/3) = 0.27$
3rd reflection $= 4(0.2)^2(1/3) = 0.053$
4th reflection $= 4(0.2)^2(1/3)^2 = 0.018$
5th reflection $= 4(0.2)^3(1/3)^2 = 0.0036$



$$V_{Z_L} = \frac{75}{100+75}(12) = 5.14V$$

Line will settle to 5.14V